claude-windows / 45ea3a91-654d-4972-9cd3-65f35db54caf

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\45ea3a91-654d-4972-9cd3-65f35db54caf\subagents\workflows\wf_2a64bc88-2a8\agent-a9fbdbe4d15ef40c3.jsonl`
- SHA-256: `3a6eef3a04a7dc799d5abd739afa6b7110cf33142f1a1bbbabae31504cd2e38d`
- Source modified: `2026-06-10T06:10:48+00:00`
- Imported at: `2026-07-05T16:48:27+00:00`
- project: `wf_2a64bc88-2a8`
- session_id: `45ea3a91-654d-4972-9cd3-65f35db54caf`

Transcript

1. user (2026-06-10T06:09:07.935Z)

```
CONTEXT ? two sibling repos on a Windows machine:
- INDEXER: C:/Users/avidu/Projects/badminton-highlight-indexer ? local AI badminton(->racket-sports) highlight indexer + rally detector. FastAPI + SQLite + ffmpeg + GPU CV via WSL; Gemini = paid cloud AI. docs/ tree is rich. Current branch docs/next-steps-multisport (master = main branch).
- COLLECTOR: C:/Users/avidu/Projects/sports-data-collector ? golden-label acquisition/curation/ingestion CLI (system-of-record candidate for training corpus).
Project state (June 2026): scaffolding complete; owned-model roadmap agreed (docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md) but NO platform/owned-model implementation started; next committed platform slice = async job model. Licensing guardrail: MIT/Apache-2.0/BSD only for code+data+weights; paid LLMs are inference-only, never training-data sources.
ENVIRONMENT RULES: this dev environment has NO GPU/torch/cv2 ? do NOT attempt to run the video pipeline or import torch/cv2 modules. Test suites are offline/fully-mocked and SAFE to run. You are READ-ONLY: never edit/create/delete repo files; running tests and git read commands is allowed.
QUALITY BAR: cite file paths (and line numbers where possible) for every finding. Report what IS in the code/docs, not what you assume. Be skeptical and concrete. Your final structured output is consumed programmatically by an orchestrator ? return raw data, not prose for a human.
ROLE: adversarial verifier. An auditor ("docs-strategy") claimed the finding below. Try to REFUTE it against the actual files. Default to isReal=false if the evidence does not hold up, the behavior is documented/deliberate (e.g. listed in docs/CODE_MAP.md gotchas or DEFERRED_HARDENING_PLAN.md as a known deferral ? unless the auditor adds genuinely NEW consequence), or the severity is materially overstated.
FINDING: {"title":"Gemma-4 stance contradiction: candidate clean base (CLAUDE.md/PLATFORM/COMMERCIALIZATION) vs AMBER/bench-only requiring counsel (plan/study banner)","severity":"high","area":"licensing guardrails / owned-model base","file":"C:/Users/avidu/Projects/badminton-highlight-indexer/docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md:42","evidence":"CLAUDE.md:34 lists Gemma 4 as an accepted base: 'Owned-model base: Apache-2.0, video-capable (Qwen-VL / Gemma 4)'. docs/PLATFORM_ARCHITECTURE.md:304: 'Gemma 4 multimodal (Apache-2.0, 2026) is a clean Google-stack alternative to evaluate' and ~line 313: 'Gemma 4's switch to Apache-2.0 removes this'. docs/COMMERCIALIZATION.md C14 (line 43) includes 'or Gemma 4' in the approved-bases list. Versus docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md:42-43: 'Gemma-4 stays AMBER and bench-only ? its Apache grant is likely but its Prohibited-Use posture is unconfirmed (counsel needed), and its video/timestamp capability is disputed; not the primary extractor', and the study's correction banner (docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md:22-23): '? Gemma 4 stays AMBER and bench-only (NOT \"confirmed clean\" as the table below says)'. NEXT_STEPS.md:11 agrees: '(Gemma-4 AMBER/bench)'.","why_it_matters":"A reader of CLAUDE.md or PLATFORM would treat Gemma 4 as a cleared, Apache-confirmed base and could build on it without the counsel check the newer (authoritative) plan requires ? exactly the class of license mistake the Gemma-3 rejection was meant to prevent."}
Steps: open the cited file(s) and any counter-evidence files; verify quotes/line claims; judge severity honestly. Cheap greps allowed; do NOT run the pipeline. Return isReal, confidence, and
```

a 1-3 sentence verdict (with corrected detail if partially right).

2. user (2026-06-10T06:09:23.933Z)

```
CLAUDE.md:34:- **Owned-model base:** Apache-2.0, video-capable (Qwen-VL / Gemma 4); **reject
Gemma 3**
docs\PLATFORM_ARCHITECTURE.md:260:> **The moat is *not* the model.** Anyone can download Qwen-
VL/Gemma. The defensible
docs\PLATFORM_ARCHITECTURE.md:269:> are open, Apache-2.0 models only** (Qwen-VL / InternVL /
Gemma 4). Detail in the
docs\PLATFORM_ARCHITECTURE.md:304:**Qwen3-VL-4B/8B (Apache-2.0)**; **Gemma 4 multimodal
(Apache-2.0, 2026)** is a clean
docs\PLATFORM_ARCHITECTURE.md:309:- ? **Reject for the flagship:** **Gemma 3** ? custom terms
with a **viral derivative
docs\PLATFORM_ARCHITECTURE.md:311:  permanently bound to Google's *updatable* Prohibited-Use
Policy); **Gemma 4's switch to
docs\PLATFORM_ARCHITECTURE.md:347:| **Together AI** | Managed (tune + serve) | Qwen3-VL /
Gemma-3 / Llama-4; image data | verify | partial | ? adapters downloadable | managed Qwen3-VL +
serving |
docs\PLATFORM_ARCHITECTURE.md:419:InternVL / Gemma 4 ? Apache-clean); **(2)** keep the paid API
strictly as a **runtime
docs\PLATFORM_ARCHITECTURE.md:531:   (Qwen-VL Apache-2.0 vs Gemma 3 custom license vs ?) and
confirm a gated commercial
docs\PLATFORM_ARCHITECTURE.md:563:[Omitted long matching line]
docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md:25:  (Ratifies the prior "Apache-2.0 only" to include
MIT/BSD.) Reject viral/NC/custom-restrictive: Gemma 3 (viral),
docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md:26:  **Gemma 3n** (Gemma Terms follow derivatives ? even
cleaner reject), Llama 4 (700M-MAU cap), Commons-Clause repos.
docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md:35:**Framework ? base** (the Gemma-4-vs-ms-swift
clarification): a *framework* is the forge (runs SFT/GRPO, emits
docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md:42:  pre-release. **Gemma-4 stays AMBER and bench-only**
? its Apache grant is likely but its Prohibited-Use posture is
docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md:93:  target stays reachable without re-architecting.
(On-device base stays on the Qwen3-VL-4B Apache lineage ? *not* Gemma-3-270M.)
docs\archives\checkpoints\2026-06-strategy-foundation\README.md:25:- **Owned-model base =
Apache-2.0-clean, video-capable** (Qwen2.5-VL-7B/32B, Qwen3-VL, Gemma 4); **reject Gemma 3**
(viral license clause); **minors excluded** from the training pool; per-user training needs a
separate opt-in.
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:22:- ? **Gemma 4 stays AMBER and bench-
only** (NOT "confirmed clean" as the table below says): Apache grant likely but
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:37:| **Gen-2 / Option B** | 12?24+ mo,
after scaffolding + healthy corpus | end-to-end multimodal VLM: (frames+audio+pose+trajectory) ?
rally [start,end] (+semantics); SFT cold-start then **GRPO/RFT with a temporal-IoU-vs-golden
reward** | **Qwen3-VL-4B-Instruct** (Apache-2.0, on-device) primary; **Qwen2.5-VL-7B** fallback;
**Gemma 4 E4B/12B-Unified** (Apache-2.0 + native video+audio) end-state | rented 24?48 GB
A100/L40S (does NOT fit 8 GB box) |
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:98:  Gemma 4 E2B/E4B/12B-Unified
(Apache-2.0 + native video/audio), InternVL3 ?38B (MIT/Apache), ASFormer/
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:99:  ActionFormer/TriDet (MIT).
**REJECTED:** Gemma 3 (viral Gemma Terms + PUP), Qwen2.5-VL-3B (NC research),
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:102:  (Commons Clause ? use `sj-li/MS-TCN2`
MIT). ? **Gemma 4 size names = E2B/E4B/26B-A4B/31B/12B-Unified ? there is no plain "4B".**
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:127:- **Gemma 4 "clean + native-video"
rests on post-cutoff secondary sources** ? keep AMBER until a raw HF LICENSE
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:128:  revision is verified, given the
Gemma-3 rejection. Its timestamp-emission recipe is less documented than Qwen's.
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:147:3. Working Gemma-4 timestamp-emission
recipe + exact checkpoint (E4B vs 12B-Unified)?
docs\NEXT_STEPS.md:11:rally-vs-non-rally problem). **Framework = ms-swift, base = Qwen3-VL**
(Gemma-4 AMBER/bench) come **later** for the
docs\COMMERCIALIZATION.md:43:[Omitted long matching line]
```

3. user (2026-06-10T06:09:25.193Z)

```
1       # Owned-model implementation plan (roadmap) ? DRAFT for owner review
```

Status: PLAN / roadmap. No owned-model *implementation* starts until the owner greenlights a specific
milestone. Paid Gemini stays the always-ready inference/fallback throughout (never a training source).
Grounded in: `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md`, the n=6 LOVO + learned prototype
(`docs/archives/quality-iterations/2026-06-multivideo-lovo/`), and a verified strategy workflow (2026-06-09).

Design principle (owner directive): *envision now, build later*
Every deferred piece (conversational-QA, on-device, multisport, the VLM) must be **architecturally envisioned
now** so we never hit a mammoth refactor. We **design the seams** (data contracts, export boundaries, the
event store) up front and **implement behind them** when each milestone is greenlit. Punting a *feature* is fine;
punting its *seam* is not.

North Star (reframed per PR #61 review)
- **Immediate deliverable:** a **training framework + harness** that produces a model with **very high precision/
  recall for extracting highlights + rallies from a clip**. *Quality is the first-order objective.*
- **Eventual product:** a **single, sport-agnostic, conversational** video model ? upload a clip, then ask
  contextual questions ("longest rally", "make a clip of all service faults", "rallies over 20 shots").
- **On-device is LATER** (a model-compression goal so power users save upload cost/hassle) ? definitely on the
  roadmap, but **not** the immediate target. Don't over-index the strategy on it now.
- **The moat = the consented dataset + the eval harness, not the weights.**

Hard guardrails (non-negotiable ? "never corrupted")
- **Commercial-clean licensing for EVERY dependency ? code, data, AND model weights: MIT / Apache-2.0 / BSD only.**
  (Ratifies the prior "Apache-2.0 only" to include MIT/BSD.) Reject viral/NC/custom-restrictive: Gemma 3 (viral),
  Gemma 3n (Gemma Terms follow derivatives ? even cleaner reject), Llama 4 (700M-MAU cap), Commons-Clause repos.
- **Never train on paid-LLM (Gemini/OpenAI/Anthropic) outputs.** Also forbidden: public grounding-CoT datasets that
  were Gemini-generated (e.g. TVG-R1's 56K) ? we must originate our own CoT supervision (a real, binding cost).
- **Exclude minors; per-user training data needs a separate explicit opt-in; split data by match.**
- ? **Base-model pretraining provenance is unauditable for *every* open VLM (incl. Qwen3-VL).** Our "no paid-LLM
  training" rule is enforceable at *our fine-tuning data* layer, not the base's pretraining ? an **explicit owner
  risk-acceptance** is required to use any open VLM. (Governed by the base's license, not our pipeline.)

The model strategy (verified 2026-06-09)
Framework ? base (the Gemma-4-vs-ms-swift clarification): a *framework* is the forge (runs SFT/GRPO, emits
weights); a *base* is the metal you forge. You pick one framework and feed it any clean base.
- **Framework: ms-swift (Apache-2.0)** ? the one forge that does native **video + audio + timestamp-grounding +
  GRPO/RFT with a custom video reward** in one stack. Keep **unsloth** (Apache-2.0 core; *avoid its AGPL Studio UI*)
  as the 8 GB-local SFT/compression tool for the deferred on-device goal. ? Point-in-time snapshot ? re-verify at impl.
- **Base (when we reach the VLM): Qwen3-VL (Apache-2.0, uniform across sizes)** ? long-

video context, purpose-built
timestamp emission (second-level localization), multi-turn video chat. Re-confirm the exact size's LICENSE
pre-release. **Gemma-4 stays AMBER and bench-only** ? its Apache grant is likely but its Prohibited-Use posture is
unconfirmed (counsel needed), *and* its video/timestamp capability is disputed; not the primary extractor.
InternVL3 (MIT/Apache) = clean fallback if a Qwen blocker appears. ? base swap is cheap *only within
timestamp-capable Apache bases* (timestamp data-format + grounding-CoT are base-specific).
- **START WITH THE TAL HEAD, NOT THE VLM** (the decisive call): fine-tuned VLMs **miss strict temporal boundaries**
(best RL-post-trained video VLM ? R@0.7 50% on open-domain benchmarks; "coarse regions, not accurate boundaries").
Boundary accuracy IS the product. The head (`rally_seq_proto.py`) is already de-risked on our corpus (held-out
AUC 0.83?0.92), trains in minutes at ~4.5 GB on the 2070 Super ($0), is ~1M params (trivially compressible later),
and **doubles as a pseudo-label teacher + the honest baseline the VLM must beat.** So: **head now ? VLM later**,
strictly sequenced. When the VLM comes, prefer **two-stage** (VLM coarse-proposes, head refines boundaries) over
wholesale replacement. *(Open: the VLM-ceiling number is open-domain; our amateur footage may differ ? re-measure.)*

Conversational video-QA ? build the bookkeeping NOW, punt the model
Punt the conversational *feature* until extraction quality is solid (current LOVO ~0.583 ? "very high"; a chat UI
over a weak extractor exposes the weakness). **Reject end-to-end-VLM-answers-each-turn** (expensive; bad at exact
counting; the strong teachers are license-forbidden). **But build the *seam* now** (the brief requires it):
- **Architecture = "extract once, query many":** EXTRACTION (TAL head ? later VLM) emits structured timestamped
spans ? a durable **per-video EVENT STORE** ? a **structured QUERY layer** (text-to-SQL + ffmpeg clip-cut). The
cited queries ("longest rally", "all service faults", "rallies > 20 shots") are **structured-numeric ? SQL, not a
VLM job** (SQL counts exactly where VLMs hallucinate).
- **Designed-now seam:** make the **per-video event-index schema a first-class data contract** alongside the M0.1
corpus spine: per rally `{video_id(MD5), rally_ordinal, sport, start, end, shot_count, ending_reason/fault_tags,
derived_clip_ref, provenance, confidence}`. **Reserve the event/fault-tag taxonomy (e.g. `service_fault`) up front**
even if unpopulated. ? This is a real DB delta (a `PRAGMA user_version` migration): `play_segments` today has
`video_id/start_time/end_time/duration/server/receiver/metadata` only; `highlights` isn't a table yet ? so it's an
extension with new columns + a highlights table, not a no-op. **Punt** the NL parser/agent until quality clears the bar.
- Paid Gemini may answer genuinely open-ended edge questions **at inference** over our structured index (allowed ?
it reasons over our data, it is not trained on its outputs).

Multisport (single sport-agnostic model)
- **Model/framework: UNCHANGED.** One shared TAL head + one Qwen3-VL base, fine-tuned across sports (sport = an input
feature / per-sport calibration). The collector + indexer are already sport-generic. ? **ACTION: rename the repo off
"badminton" ASAP** (owner emphatic) ? consistent with this.
- **What multisport changes = DATA + DETECTORS (scale up, the accepted "good problem"):** each new sport needs its own
labeled matches (split by match) + a **clean trajectory detector** to feed the head's

features. **The binding blocker
77   is per-sport detectors:** WASB (badminton) has the unresolved **C3** checkpoint-
provenance issue, and **clean MIT/
78   Apache/BSD shuttle/ball detectors for tennis/TT/pickleball/padel are not yet
identified.** Sequence by public-data
79   richness: **badminton ? tennis/TT ? pickleball/padel.**
80   - ? **Single-model-multisport-temporal-generalization is an OPEN HYPOTHESIS**, not
proven ? our corpus is badminton-only,

4. user (2026-06-10T06:09:40.808Z)

295   golden set are already the labeling seam); **Compute/Orchestration** runs training and
296   inference jobs (even **BYO-compute** can serve inference on the tenant's GPU); the
297   **eval harness** already in `backend/eval/` extends to grade QA quality. **In other
298   words, the rock-solid scaffolding is exactly what makes the model flywheel possible** ?
299   which is why scaffolding comes first (see §6).
300
301   ### Base model & training (Apache-2.0 only ? licensing is decisive)
302   **Recommended near-term base: `Qwen2.5-VL-7B-Instruct`** ? Apache-2.0, native long-video
303   (dynamic-FPS sampling, temporal mRoPE), mature LoRA tooling. Upgrade path
304   **Qwen3-VL-4B/8B (Apache-2.0)**; **Gemma 4 multimodal (Apache-2.0, 2026)** is a clean
305   Google-stack alternative to evaluate.
306   - ?? **Lock to Apache-2.0-clean bases** so you can fine-tune, **keep weights private,
307   gate behind subscriptions, and never owe open-sourcing or branding**. (Note Qwen2.5-VL
308   is Apache **only at 7B/32B** ? 3B/72B use the custom Qwen license.)
309   - ? **Reject for the flagship:** **Gemma 3** ? custom terms with a **viral derivative
310   clause** (anything fine-tuned on it, *or even trained on its synthetic outputs*, stays
311   permanently bound to Google's *updatable* Prohibited-Use Policy); **Gemma 4's switch
to
312   Apache-2.0 removes this**. **Llama 3.2 Vision** ? custom license forces a "Llama-?"
313   model name + "Built with Llama" badge and a 700M-MAU ceiling. **`gpt-oss`** ?
Apache-2.0
314   but **text-only**, so usable only as an optional text-reasoning layer, never the video
315   encoder.
316   - **Training method:** **QLoRA** (? r=64, ?=128) on the 7B ? fits one prosumer GPU and
317   avoids catastrophic forgetting of general video skills. Build the dataset by
converting
318   the existing rally/shot/segment annotations + golden set into templated **clip?QA
319   pairs**, expand synthetically with **open-model teachers only** (see constraint
below),
320   and hold out a **golden QA eval** that gates every fine-tune. A few hundred gold pairs
?
321   **5k?20k** curated is a realistic band for this narrow domain.
322   - **Serving economics:** a 7B VLM ? 14 GB VRAM (FP16) / ~3.5 GB (INT4); a ?16 GB GPU is
a
323   comfortable serve target (vLLM for throughput). Self-hosting wins **only** once you
have
324   sustained video-QA volume **+** a defensible consented dataset **+** real per-call API
325   cost pain ? below that, the paid API is cheaper than a standing GPU and the second
stack
326   is overhead. Hence: **paid-primary now ? owned-primary / paid-fallback later.**
327
328   Base-model licensing + training-data policy are tracked as **COMMERCIALIZATION C14**.
329

5. user (2026-06-10T06:09:41.763Z)

35 | C5 | **Output codec royalties** | **P0** | ? | Highlights are still encoded **H.264 +
AAC** (`libx264` in `backend/pipeline/stitcher/compiler.py`) ? encode-side AVC patent exposure
remains. | Switch output to **AV1 + Opus** (royalty-free) on an **Ada-class GPU (L4/L40)** for
HW encode. See §4. Decision: provision Ada-class GPU? |
36 | C9 | **Packaging / distribution** | **P0** | ? | No `pyproject.toml`; the diagnostic
`setup.py` shadows the build system. | Design in `DEFERRED_HARDENING_PLAN.md` (#7) ? **awaiting
your approval** before code. |
37 | C12 | **SaaS foundations (async/auth/tenancy)** | **P1** | ? | Sync `/api/process`

blocks on long videos; no auth/tenancy. | Multi-user/platform strategy in
`PLATFORM_ARCHITECTURE.md` (pluggable storage/compute, tenancy, orchestration); async slice =
`DEFERRED_HARDENING_PLAN.md` (#16). |
38 | | **? IN-FLIGHT (in progress / watch)** | | | | |
39 | C4 | **Runtime AI strategy (A/B/C + owned model)** | **P0** | ? | Live = **(C) Gemini
2.5 Flash, paid tier**. Long-term = **own a self-hosted generative video-QA model** (fine-tuned
open VLM) as primary, **paid Gemini always-ready as fallback** via the provider registry. | Make
"provider registry + paid fallback" a core principle; build the owned model **after** the
scaffolding/flywheel are solid. See `PLATFORM_ARCHITECTURE.md` §5A. |
40 | C13 | **Beachhead validation** | **P0** | ? | Market research
(`PMF_MARKET_RESEARCH.md`) recommends **India badminton academies** as the #1 beachhead.
Unvalidated. | Cheapest tests: 10?15 academy/coach discovery interviews; rally-detection P/R
benchmark on 100+ real amateur clips; fake-door pricing (B2C ~$99/yr, B2B academy ~$600/yr). See
§6. |
41 | C8 | **Golden-set vendoring spend** | **P1** | ? | ADR-006: **Roora (Bengaluru)**
selected for an initial paid pilot; licensed/owned video only (no user uploads to vendors). |
Confirm quality-per-dollar from the pilot; then GS-3 `HumanVendorProvider`. See
`GOLDEN_SET_VENDORING.md`. |
42 | C3 | **Shuttle-tracking weights provenance** | **P1** | ? ?? | WASB-SBDT **code is MIT
(clear)**; distributed **checkpoints are academic-data-trained** ? not covered by the MIT code
grant. MonoTrack/YOLO-NAS banned (noncommercial). | Train-own weights (Phase-4 roadmap) **or**
confirm per-checkpoint terms ?? before commercial deploy. |
43 | C14 | **Owned-model base licensing + training-data policy** | **P1** | ? ?? | Owned
video-QA model (PLATFORM §5A) must build on **Apache-2.0-clean, video-capable** bases ?
Qwen2.5-VL-7B/32B (Apache; 3B/72B are *not*), **Qwen3-VL-4B/8B**, or **Gemma 4** ? to keep
weights private, gate, and monetize. **Reject Gemma 3** (viral derivative clause ? permanent
Google PUP), Llama Vision (naming + 700M-MAU), `gpt-oss` (text-only). **Hard rule: never train
on paid-LLM (Gemini/OpenAI/Anthropic) outputs** (competing-model clause). The moat is the
consented dataset + eval harness, not the model. | Lock to Apache-2.0; **open-model teachers
only**; separate explicit **training opt-in**; **exclude minors** from the training pool; per-
clip consent ledger. See PLATFORM §5A/§7. |
44 | C6 | **Inbound HEVC decode** | **P2** | ? ?? | Decoding users' GoPro H.264/**HEVC**
uploads is unavoidable; HEVC pools are fragmented + highest-uncertainty. | Low/uncertain
residual; confirm decode-side exposure with counsel before launch. |
45 | C7 | **Hosting unit-economics** | **P2** | ? | Per-video cost is *cheap* for Gemini
(~$0.05 input for a 10-min clip); risk is volume × latency, not unit price. Self-host/offline
costs unmeasured. | **Benchmark $/video & latency on the target VM** for A/B/C before pricing.
See §5. |
46 | C10 | **Value layer (product surface)** | **P2** | ? | Minimal exports shipped
(EDL/JSON/CSV + `basic_stats`, PR #51). | Deepen: per-rally tags/notes that feed golden, per-
player/match stats, clip gallery. See §6. |
47 | C11 | **Input-quality enforcement** | **P2** | ? | Per-video observability reports
surface a customer verdict + footgun flags incl. audio-present (PR #50). | Turn more
`VIDEO_RECORDING_GUIDELINES` dimensions into validators/report rules. |
48 | C2 | **`supervision` ByteTrack pin** | **P2** | ? | Pinned `<0.30.0` (deprecation
ahead). | Migrate before lifting the pin (BACKLOG). Not a legal risk, a maintenance one. |
49 | | **? COMPLETED** | | | | |

6. user (2026-06-10T06:09:43.030Z)

1 # Owned-model training study (rally segmentation ? end-to-end multimodal)
2
3 **Date:** 2026-06-08 · **STATUS: DESIGN BLUEPRINT (no implementation until PLATFORM
P0?P4 + healthy corpus; owner sign-off required).**
4
5 > How to train our **OWNED** rally/highlight model **reusing the videos we label**,
aligned to the
6 > Apache-2.0 / no-paid-LLM-training / privacy-local guardrails. Produced by an 11-agent
research
7 > workflow (ground ? 5 web-research dimensions ? adversarial verification of
costs/licenses/privacy ?
8 > synthesis + completeness critic; 35 sources). Empirical results from THIS session
(LOVO + a learned
9 > prototype) are folded in and **resolve the study's biggest open unknown** ? see §"What
we proved today".

10 > Numbers from web sources are 2026 snapshots in a ±15?25% band; **re-quote live before budgeting.**
11
12 ## Update (2026-06-09, post-PR-#61-review) ? authoritative roadmap is now `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md`
13 A verified strategy workflow + owner review refined this study. Key changes (the plan supersedes where they differ):
14 - **VISION (foregrounded):** the goal is **rally/shot/player analytics + highlights for ALL net-separated racquet
15 sports** (badminton/tennis/TT/pickleball/padel) ? eventually all sports, as **ONE sport-agnostic, conversational
16 model**. This does **not** change the model/framework picks; it scales **data + per-sport trajectory detectors** (the
17 accepted "good problem") and adds a licensing front (clean per-sport detectors). Single-model temporal generalization
18 is an **open hypothesis** (corpus is badminton-only; the once-cited RacketVision evidence was **refuted** ? it's
19 ball-tracking, not temporal).
20 - **Framework = ms-swift (Apache-2.0)** (only forge with native video+audio+timestamp-grounding+video-GRPO).
21 **Base = Qwen3-VL (Apache-2.0, uniform sizes, timestamp emission)** ? primary, **replacing the Qwen2.5-VL-7B framing**.
22 - ? **Gemma 4 stays AMBER and bench-only** (NOT "confirmed clean" as the table below says): Apache grant likely but
23 Prohibited-Use posture unconfirmed (counsel) + video/timestamp capability disputed ? not the primary extractor.
24 - **Start with the TAL head, not the VLM** (VLMs miss strict boundaries ? boundary accuracy is the product). Head now ? VLM later.
25 - **Immediate goal = a high-P/R extraction *harness*, not on-device** (on-device deferred). **Reframe supersedes the on-device-now framing.**
26
27 ## Thesis
28
29 Train the owned model in **three staged generations off ONE model-agnostic labeled corpus** ? a label
30 collected once feeds every generation, so labels are never re-collected. **The moat is the consented
31 dataset + the eval harness, never the weights.**
32
33 | Gen | When | Approach | Base / model | Compute |
34 |---|---|---|---|---|
35 | **Gen-0** | now ? unblocks at n?5 matches | tiny **temporal action segmentation (TAS)** head over a hand-built per-frame feature stack from the WASB trajectory we already produce; per-frame rally(1)/non-rally(0) ? merge into [start,end] | **ASFormer (MIT, ~1.1M params)** primary; **ActionFormer (MIT)** A/B challenger (overtakes once corpus = hundreds of matches) | **local RTX 2070 Super**, trains in minutes, $0 |
36 | **Gen-1 / Option A** | 6?12 mo | concatenate more local cue channels onto the SAME head: trajectory + audio-onset + pose/court serve-gating; fusion in our rally layer, cues swappable | same ASFormer/ActionFormer, wider input | local + occasional cloud burst (pseudo-labeling) |
37 | **Gen-2 / Option B** | 12?24+ mo, after scaffolding + healthy corpus | end-to-end multimodal VLM: (frames+audio+pose+trajectory) ? rally [start,end] (+semantics); SFT cold-start then **GRPO/RFT with a temporal-IoU-vs-golden reward** | **Qwen3-VL-4B-Instruct** (Apache-2.0, on-device) primary; **Qwen2.5-VL-7B** fallback; **Gemma 4 E4B/12B-Unified** (Apache-2.0 + native video+audio) end-state | rented 24?48 GB A100/L40S (does NOT fit 8 GB box) |
38
39 ## What we proved today (de-risks the blueprint)
40

7. user (2026-06-10T06:09:43.627Z)

1 # Next steps (live) ? owned-model + multisport
2
3 **Updated:** 2026-06-09 (end of the quality-iteration + multisport-recon session). **Authoritative roadmap:**

```
4	`docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md`. **Latest blow-by-blow:** memory `current-state.md` / `session-handoff.md`.
5	This file = the prioritized "what to do next," refreshed each session.
6
7	## The recommendation (one paragraph)
8	Build the **quality-first extraction harness** before anything else: a learned **TAL head** (ActionFormer/ASFormer, MIT)
9	over trajectory features is the robust path ? it **decouples from the multi-court "background games" confound** that the
10	unsupervised heuristic (and audio, and spatial court-gating) can't escape (all three tested + failed; it's a *temporal*
11	rally-vs-non-rally problem). **Framework = ms-swift, base = Qwen3-VL** (Gemma-4 AMBER/bench) come **later** for the
12	conversational/grounding upgrade ? start with the head. **On-device is deferred** (export-clean ops now so it stays
13	reachable). Everything stays **MIT/Apache-2.0/BSD** (code + data + weights). Bottleneck = **data + per-sport trajectory
14	detectors**, not model architecture.
15
16	## Latest findings (2026-06-09)
17	- **6-match badminton LOVO:** learned prototype **0.583** vs heuristic **0.480** (learned leads since n=3, AUC 0.83?0.92,
18	  decoupled from multi-court). The "contrast" metric (in-rally÷dead-time visibility) predicts heuristic F1 monotonically.
19	- **?? Multisport (Roora pickleball + TT):** labels **ingest cleanly** (`curate convert-cvat` ? 6 rallies/sport, sane).
20	  **WASB-transfer SURPRISE (directional, n=6 each ? TINY):** the badminton shuttle detector segments **TABLE TENNIS at
21	  F1 0.909** out-of-the-box (!) but **pickleball only 0.308**. ? **The per-sport-detector blocker is SPORT-DEPENDENT:** TT may
22	  be servable with WASB short-term; pickleball needs a proper (MIT/Apache) ball detector. Confirm/deny with more videos.
23	- **Owner field notes (2026-06-09):** camera height/distance **varied on purpose** (great generalization data); **multi-court
24	  adjacent-court SOUND dominates the signal** ? empirically reconfirms audio-foreground isolation won't work + the
25	  robustness-first learned-model path. Multi-court is the realistic (academy) environment, so this *is* the target distribution.
```

8. user (2026-06-10T06:10:11.747Z)

```
docs\COMMERCIALIZATION.md:12:**Merged from / supersedes:**
docs\COMMERCIALIZATION.md:218:- `archives/COMMERCIAL_READINESS_REVIEW.md` ? superseded assessment (history).
docs\GOLDEN_SET_IMPLEMENTATION.md:58:An independent strong labeler for the fast/CI tier. **Must be a different model lineage than whatever production uses** (else shared blind spots ? false green). A fast-tier failure is a *smoke alarm* that escalates to the GS-3 human tier, never an authoritative verdict on its own.
docs\archives\COMMERCIAL_READINESS_REVIEW.md:3:> **STATUS: SUPERSEDED (archived 2026-06-07).** The commercial findings and the
docs\GOLDEN_SET_VENDORING.md:94:| `HumanVendorProvider` | The bake-off winner; authoritative ground truth, corpus growth, deep audits | **Slow (days)** |
docs\NEXT_STEPS.md:3:**Updated:** 2026-06-09 (end of the quality-iteration + multisport-recon session). **Authoritative roadmap:**
docs\NEXT_STEPS.md:11:rally-vs-non-rally problem). **Framework = ms-swift, base = Qwen3-VL** (Gemma-4 AMBER/bench) come **later** for the
docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md:42:  pre-release. **Gemma-4 stays AMBER and bench-only** ? its Apache grant is likely but its Prohibited-Use posture is
docs\README.md:7:- [COMMERCIALIZATION.md](COMMERCIALIZATION.md) ? Living tracker for the commercial dimension (licensing, codec/output economics, hosting, vendoring, packaging) + PMF risks and the quality/data-phase priorities. Supersedes the archived `COMMERCIAL_READINESS_REVIEW.md`.
docs\ROADMAP.md:78:verdict **AMBER / leaning-negative**: recall signal is real (hit-clusters recover
docs\ROADMAP.md:108:| + | Audio hit-detection cue (`backend/pipeline/audio/`, ADR-007) | ? E1
```

```
probe run ? AMBER, gated on golden labels |
docs\archives\README.md:12:- `research/` ? Completed or superseded research notes and plans.
docs\archives\decisions\DECISIONS.md:273:3. **Pluggable `AnnotationProvider` registry**
mirroring the existing registry pattern: `FileProvider` (today's manual/ShuttleSet flow),
`HumanVendorProvider` (slow tier ? authoritative truth), `AutomatedProvider` (fast tier ?
independent model oracle).
docs\archives\research\AUDIO_RALLY_DETECTION_PLAN.md:97:TestLargeVideo (7). **Verdict: AMBER ?
leaning negative for the *classical* detector.**
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:12:## Update (2026-06-09, post-
PR-#61-review) ? authoritative roadmap is now `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md`
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:13:A verified strategy workflow + owner
review refined this study. Key changes (the plan supersedes where they differ):
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:22:- ? **Gemma 4 stays AMBER and bench-
only** (NOT "confirmed clean" as the table below says): Apache grant likely but
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:25:- **Immediate goal = a high-P/R
extraction *harness*, not on-device** (on-device deferred). **Reframe supersedes the on-device-
now framing.**
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:127:- **Gemma 4 "clean + native-video"
rests on post-cutoff secondary sources** ? keep AMBER until a raw HF LICENSE
docs\archives\quality-iterations\README.md:18:- **`2026-06-initial/`** ? SEED. Opened the
quality phase ahead of the awaited labeled-video batch (status: in progress; superseded for
first results by the iteration below).
```

9. user (2026-06-10T06:10:13.995Z)

```
3fd7232 2026-06-08 docs: handoff-consistency sweep + strategy checkpoint + CLAUDE.md (#59)
757842c 2026-06-08 docs(platform): address review ? v1-vs-vision scope, Hatchet framing, BYO ops
burden
8fe780e 2026-06-08 docs(commercialization): group dashboard Pending -> In-flight -> Completed +
add Priority (P0/P1/P2) column
4d04a73 2026-06-09 Quality iteration: human-GT multi-video LOVO + learned prototype + multi-
court finding + owned-model study/plan (#61)
```

10. user (2026-06-10T06:10:43.404Z)

```
Structured output provided successfully
```

11. assistant (2026-06-10T06:10:48.404Z)

```
Verification complete.
```